

4	(a) (i) stress = force / cross-sectional area	B1	[1]
	(ii) strain = extension / <u>original</u> length	B1	[1]
	(b) (i) $E = \text{stress} / \text{strain}$	C1	
	$E = 0.17 \times 10^{12}$	C1	
	stress = $0.17 \times 10^{12} \times 0.095 / 100$	C1	
	= $1.6(2) \times 10^8 \text{ Pa}$	A1	[4]
	(ii) force = (stress $\times$ area) = $1.615 \times 10^8 \times 0.18 \times 10^{-6}$	C1	
	= 29(.1) N	A1	[2]